

Computational Social Science with Images and Audio

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To move from images to audio, let us see what Dall-E 2 says about audio analysis and human speech.



Figure: "The auditory sense of humans is important, symbolized"



Figure: "Artwork on human speech"

Human hearing is interesting (not only for social scientists).

- ▶ Hearing is a critical sense for communication and survival
- ▶ As for vision, it enables us to interact with the world without physical contact
- ▶ Various areas of the brain are associated with auditory processing
- ▶ Humans are “analysts” and “generators” of audio at the same time
- ▶ Human hearing has its constraints
 - ▶ Limited frequency range: Humans typically hear between 20 Hz to 20 kHz, whereas some animals can perceive sounds beyond these limits
 - ▶ Difficulty in locating low-frequency sounds
- ▶ Human hearing is fascinating on several levels:
 - ▶ Fundamental: How does the auditory system decode sounds?
→ Beyond the scope of this class
 - ▶ Aggregate/social: What is the impact of sound on societal outcomes? → Central to our class

Audio analysis comes with many definitions.

- ▶ Many view it as a signal-processing challenge
- ▶ Julius O. Smith III: reconstructing auditory experiences
- ▶ In this class, we will explore techniques to analyze auditory data (recordings) in large quantities to study social science questions
 - ▶ Again, we begin where the auditory data already exists: we do not cover, from a technical perspective, the mechanisms of capturing this data (e.g., microphone technology)

What is sound?

- ▶ A sensory experience created by vibrations traveling through a medium (usually air)
- ▶ Like ocean waves move through water, sound waves propagate through the air
- ▶ As ocean waves interact with air, they create ripples in the atmosphere, analogous to **sound waves**
 - ▶ We experience these as the ambient sounds of the ocean

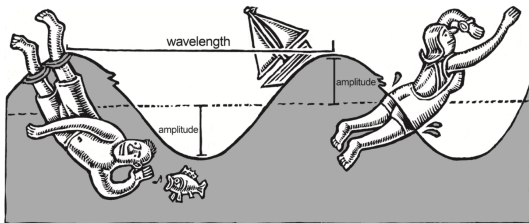


Figure: The analogy of ocean waves to sound waves; Source: Sulzer, (2021)¹

¹Sulzer (2021). Music, math, and mind: The Physics and Neuroscience of Music. Columbia University Press.

Humans' hearing range is 20 Hz to 20K Hz.

- ▶ We begin to hear low vibrations at around **20 Hz**
- ▶ Teenagers can hear frequencies that extend up to about **20,000 Hz**, above which we humans do not perceive sound
- ▶ Each time you double the frequency, you move up by one octave
- ▶ Hence, with good hearing, we can hear across a ten-octave range:

1. 20 Hz
2. 40 Hz ($20 \text{ Hz} \times 2$)
3. 80 Hz ($40 \text{ Hz} \times 2$)
4. 160 Hz ($80 \text{ Hz} \times 2$)
5. 320 Hz ($160 \text{ Hz} \times 2$)
6. 640 Hz ($320 \text{ Hz} \times 2$)
7. 1280 Hz ($640 \text{ Hz} \times 2$)
8. 2560 Hz ($1280 \text{ Hz} \times 2$)
9. 5120 Hz ($2560 \text{ Hz} \times 2$)
10. 10240 Hz ($5120 \text{ Hz} \times 2$)
11. 20480 Hz ($10240 \text{ Hz} \times 2$)

Figure: Human hearing range

Hz, octave, and sampling frequency are important concepts around sound waves.

- ▶ Hz (Hertz): Measures the number of cycles per second of a periodic waveform
 - ▶ Higher Hertz means a higher-pitched sound
 - ▶ If a tone has a frequency of 440 Hz, the sound wave completes 440 cycles every second (A4)
- ▶ Octave: The interval between one musical pitch and another with half or double its frequency
 - ▶ 440 Hz and 880 Hz
- ▶ Sampling frequency: The number of samples per second from a continuous signal to make a digital signal
 - ▶ Standard CD sampling rate is 44.1 kHz...
 - ▶ ... meaning that the audio waveform is being sampled 44,100 times per second

The wavelength is an important sound wave characteristic.

- ▶ What is a sound wave, to begin with?
 - ▶ A transfer of sound energy through a medium
 - ▶ It is transmitted via the vibration of particles within that medium
- ▶ Illustration for non-physicists:
 - ▶ When something vibrates (like a speaker's diaphragm), it pushes on the neighboring air particles, increasing the pressure in that region → this push is then transferred from particle to particle → when it reaches someone's ear, the ear interprets the varying pressures as sound
- ▶ Wavelength: The physical distance between identical points in consecutive cycles of a sound wave
 - ▶ Determines the pitch

The amplitude is another important characteristic.

- ▶ Amplitude *in a general wave context*: the magnitude of change in the oscillating variable within the wave
 - ▶ → Larger amplitudes mean more energy and often translate to louder sounds when talking about sound waves
- ▶ Amplitude *in digital audio signal processing*: understood as instantaneous amplitudes that represent the “magnitude of change” in the pressure wave (the sound wave) from its equilibrium position at that particular moment
 - ▶ → The specific value of the audio waveform at a given sample point
 - ▶ What is the equilibrium position?

What does this sound like?

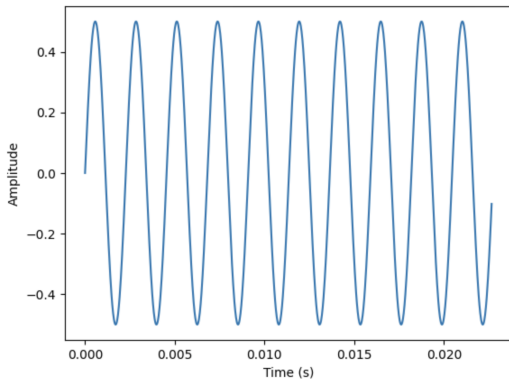


Figure: What sound could this be?

And what could this be?

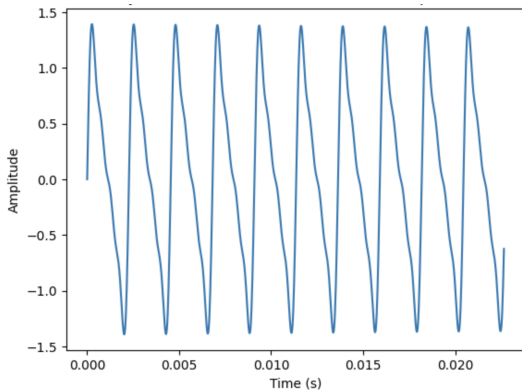


Figure: Any ideas?

This sound wave already looks more complex:

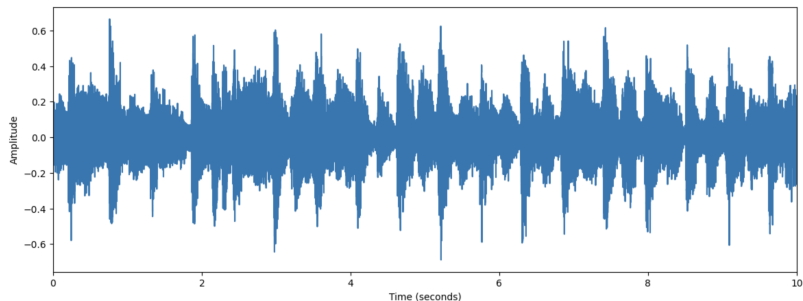


Figure: Hint: You certainly know this sound (too well).